

NetGuard AI

Intelligent Automated Network Change Management
& Self-Healing Platform

DOCUMENT VERSION

1.0

PROJECT TYPE

AI + Network Automation + DevOps

TARGET

Hackathon Prototype

CLASSIFICATION

Confidential — Internal Review

Table of Contents

1.	Executive Summary	3
2.	Problem Statement	3
3.	Proposed Solution & Workflow	4
4.	Objectives	5
5.	Target Users	5
6.	Key Features	6
7.	Functional Requirements	10
8.	Non-Functional Requirements	11
9.	Proposed Technology Stack	12
10.	Innovation Highlights	13
11.	Expected Impact	13

1. Executive Summary

Network configuration mistakes are responsible for a significant percentage of enterprise outages. Even experienced engineers can accidentally introduce configuration errors that disconnect branches, bring down routing protocols, or impact business-critical applications.

NetGuard AI is an intelligent network change management platform that automates the entire configuration deployment lifecycle — from change proposal to deployment verification and automatic recovery.

Unlike traditional change management tools that simply push configurations, NetGuard AI continuously validates changes, predicts deployment risks, monitors network health in real time, and automatically restores the previous working configuration if failures are detected.

The platform dramatically reduces Mean Time To Recovery (MTTR), prevents configuration drift, and provides enterprise-grade auditability for compliance.

2. Problem Statement

Modern enterprise networks still rely heavily on manual configuration changes. Current challenges include:

- Human errors during configuration
- No automatic rollback mechanism
- Lack of centralized approvals
- Configuration drift
- Poor visibility into deployment history
- Long outage recovery times
- No automated health validation
- Compliance issues due to missing audit trails

Risk: A single incorrect command can disconnect an entire branch office or data center, resulting in downtime and financial losses.

3. Proposed Solution & Workflow

NetGuard AI introduces an automated deployment pipeline inspired by CI/CD principles used in software engineering. Every configuration change follows an intelligent workflow:

1	Engineer initiates a Change Request
2	AI Risk Analysis evaluates potential impact
3	Syntax Validation checks command correctness
4	Network Sanity Checks confirm topology integrity
5	Manager Approval authorizes the change
6	Automatic Snapshot backs up current state
7	Configuration Deployment executes the change
8	Real-Time Health Monitoring observes post-deploy status

Healthy	→ Deployment Success
Failure	→ Automatic Rollback triggered

4. Objectives

- Prevent network outages caused by configuration mistakes.
- Fully automate configuration deployment.
- Reduce downtime through self-healing rollback.
- Provide enterprise-grade change tracking.
- Improve deployment confidence using AI-assisted validation.
- Simplify compliance and auditing.

5. Target Users

Role	Responsibilities
Network Engineers	Submit changes, review configuration differences, deploy configurations.
NOC Engineers	Monitor deployments, receive failure alerts, track health status.
Network Administrators	Approve requests, manage inventory, configure policies.
Security Teams	Review audit logs, verify compliance, investigate incidents.
Enterprise Auditors	Generate compliance reports, verify change history.

6. Key Features

6.1 Smart Change Request Portal

Every deployment begins with a structured change request. Each request includes:

- Device
- Priority
- Change description
- Maintenance window
- Proposed configuration
- Auto-generated config diff
- Business justification

6.2 AI Configuration Analyzer

Instead of only checking syntax, the AI engine analyzes risks such as duplicate IP detection, VLAN conflicts, routing loops, ACL conflicts, incorrect subnet masks, dangerous default routes, interface shutdown detection, BGP neighbor removal, and OSPF adjacency impact. The AI assigns a risk score to every proposed change.

Risk Score	Classification	Recommendation
15 / 100	Low Risk	Safe to Deploy
92 / 100	Critical Risk	Deployment Not Recommended

6.3 Intelligent Configuration Diff

Side-by-side comparison with color coding, line-by-line comparison, and version history.

Current Configuration	Proposed Configuration
<pre>interface Gi0/1 ip address 10.1.1.1</pre>	<pre>interface Gi0/1 ip address 10.2.2.1</pre>

6.4 Automated Validation Engine

Checks include syntax and network validation. Deployment stops immediately if validation fails.

- Syntax: invalid commands, missing parameters, unsupported features
- Network: duplicate IP addresses, gateway conflicts, missing VLANs
- Broken ACL references, invalid routing
- Interface dependency analysis

6.5 Automatic Configuration Snapshot

Before deployment, NetGuard AI captures an immutable snapshot:

- Download running configuration
- Download startup configuration
- Store encrypted backup

- Generate checksum
- Version the configuration

6.6 Zero-Touch Deployment Engine

Supports SSH, NETCONF, RESTCONF, NAPALM, and Netmiko, with the following capabilities:

- Multi-device deployment
- Parallel deployment
- Deployment queue
- Retry mechanism
- Failure isolation

6.7 Real-Time Health Monitoring

After deployment, NetGuard AI continuously monitors four dimensions:

Category	Monitored Items
Infrastructure	Ping, SSH, CPU, Memory, Interface status
Routing	BGP, OSPF, Static routes
Services	DNS, DHCP, HTTP, VPN
Traffic	Packet loss, Latency, Interface utilization

Monitoring window is configurable.

6.8 Self-Healing Rollback Engine

If health checks fail, the platform executes an automated recovery sequence, without manual intervention:

Failure Detected
Retrieve Snapshot
Reconnect
Restore Configuration
Verify Health
Notify Team

6.9 Live Deployment Dashboard

Real-time updates using WebSockets. Displays:

- Devices Online
- Active Deployments
- Failed Deployments
- Rollbacks
- Health Status
- CPU / Memory Usage
- Configuration Drift

● Recent Changes

6.10 Complete Audit Trail

Every activity is recorded. Audit logs cannot be modified.

Time	User	Action	Device	Result
10:22	Alice	Submitted CR	RTR-01	Success
10:28	Bob	Approved	RTR-01	Approved
10:30	System	Snapshot	RTR-01	Completed
10:31	System	Deployment	RTR-01	Success
10:34	System	Health Check	RTR-01	Passed

7. Functional Requirements

ID	Requirement
FR-1	User authentication with Role-Based Access Control (RBAC).
FR-2	Maintain a centralized device inventory.
FR-3	Create, edit, and approve change requests.
FR-4	Generate configuration differences (diffs).
FR-5	Run syntax validation on proposed configurations.
FR-6	Run AI-powered sanity checks and risk scoring.
FR-7	Take automatic configuration snapshots before deployment.
FR-8	Deploy configurations securely across supported protocols.
FR-9	Monitor post-deployment network health in real time.
FR-10	Automatically roll back failed deployments.
FR-11	Notify users of deployment events.
FR-12	Maintain immutable audit logs.

8. Non-Functional Requirements

Performance

- Deployment initiation in under 10 seconds
- Validation completed in under 5 seconds
- Rollback initiation in under 30 seconds
- Dashboard updates in under 2 seconds
- Support for 1,000+ managed devices

Reliability

- 99.9% platform availability
- Automatic retry on transient failures
- Fault-tolerant deployment engine

Security

- HTTPS for all client-server communication
- SSH key-based device authentication
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Encrypted credential storage
- AES-256 encrypted configuration backups

Scalability

- Multi-site enterprise deployments
- Hybrid cloud environments
- Data center networks
- ISP-scale networks

9. Proposed Technology Stack

Layer	Technologies
Frontend	React, TypeScript, Tailwind CSS, ShadCN UI, Recharts
Backend	FastAPI, Python, Celery, Redis
Network Automation	Netmiko, NAPALM, Paramiko, Nornir (optional)
Database	PostgreSQL
Monitoring	Prometheus, Grafana
Notifications	Email, Slack, Microsoft Teams
Containerization	Docker, Docker Compose

10. Innovation Highlights

- AI-assisted risk scoring before deployment.
- Automatic self-healing rollback based on live health checks.
- Git-style configuration version control for every device.
- Real-time deployment dashboard with live monitoring.
- Zero-touch network automation using Netmiko/NAPALM.
- Immutable audit logs for enterprise compliance.
- Multi-vendor support (Cisco, Juniper, Arista, Linux-based appliances).
- Policy-driven approvals with emergency change workflows.

11. Expected Impact

80%+	Reduction in configuration-related outages
Hours → Minutes	Improvement in Mean Time To Recovery (MTTR)
100%	Elimination of manual rollback processes
Higher	Deployment confidence via automated validation

NetGuard AI also provides enterprise-grade governance and compliance, positioning it as a foundational platform for modern, self-healing network operations.